

OntoGuard Boundary Proof Kit

Public Sanitized Decision Authorization Proof Brief

Protected by U.S. Patent Application 19/444,521 - Track I prioritized examination granted May 2026.

Executive summary

This kit shows OntoGuard governing a proposed AI output release before it becomes business consequence. The sample output was not authorized for autonomous release. OntoGuard escalated it to human review, withheld release, generated no-bind proof, executed controlled replay evidence, and exposed semantic explanation layers without disclosing protected implementation details.

Decision	ESCALATE
Release authorized	False
Release status	WITHHELD_PENDING_REVIEW
Routed to	HUMAN_REVIEW
Business effect	Autonomous release withheld pending review.
Route completeness	L4_ROUTE_ENFORCED_CONTROLLED_HARNESS (85.0%)
Production enforcement asserted	False

Commercial meaning

- AI Governance / AI Risk Review: identifies outputs that are risky, unsupported, review-heavy, or inadmissible before release.
- Decision Authorization Proof: proves the proposed movement was governed at the release boundary and did not form protected downstream effect when conditions failed.
- This artifact is designed to support buyer diligence and GTM conversations without exposing IP-sensitive internals.

1. Controlled runtime-seam proof

This section shows what happened when the AI output attempted to move toward downstream consequence.

Attempted movement	AI_OUTPUT_RELEASE
Control point	PRE_RELEASE_DECISION_BOUNDARY
Gate mode	CONTROLLED_INLINE_HARNESS
Gate invoked before release	True
Failed conditions	BENCHMARK_FAIL_REQUIRES_REVIEW, BELOW_AUTONOMOUS_ALLOW_THRESHOLD, HUMAN_REVIEW_ROUTE, SEMANTIC_CONSTRAINTS_GOVERNED, BENCHMARK_FAIL, UNCERTAINTY_UNRESOLVED, AUTONOMOUS_RELEASE_NOT_AUTHORIZED, DOWNSTREAM_RELIANCE_HIGH
Refused operation	AUTONOMOUS_OUTPUT_RELEASE
Authorization token issued	False
Downstream commit status	NOT_COMMITTED
Protected effect formed	False
No-bind status	NO_PROTECTED_EFFECT_FORMED

No-bind receipt

Receipt ID	receipt_92087ea328b01996
Reason	Autonomous release was withheld before downstream protected effect formation.
Receipt hash	c33c9d07cecc6ca4cc6625815d0dd9d2e4146280e9d319843b5c7ec968139868
Evidence pack hash	4a6c0c27db136cdc79801d297f22260db0bd1213f97dd44bc632cd4306c23024

Boundary note

This proves controlled harness refusal/no-bind behavior. It does not assert production route non-bypassability.

2. Route completeness, replay, and pressure tests

Route ID	llm_backend_output_release
Route inventory version	controlled_harness_routes@1.0.0
Route registered	YES
Tested	YES
Replayable	YES
Fail closed	True
Non-bypassable	CONTROLLED_HARNESS_ASSERTED
Outcome accounted	PENDING_HUMAN_REVIEW_OUTCOME
Limitations	OUTCOME_CLOSURE_PENDING

Replay evidence

Same-condition replay	PASSED / passed=True
Changed-condition replay	CONTROLLED_FIXTURE_PASSED / passed=True
Changed-condition reason	Changed-condition controlled fixture rejects missing/invalid authorization before downstream commit.

Route pressure tests

Test ID	Expected	Observed	Passed
missing_authorization_token_rejected	FAIL_CLOSED	FAIL_CLOSED	True
invalid_authorization_token_rejected	FAIL_CLOSED	FAIL_CLOSED	True
timeout_fail_closed	FAIL_CLOSED	FAIL_CLOSED	True
direct_downstream_commit_blocked	FAIL_CLOSED	FAIL_CLOSED	True
stale_policy_snapshot_rejected_or_escalated	FAIL_CLOSED	FAIL_CLOSED	True
mismatched_route_token_rejected	FAIL_CLOSED	FAIL_CLOSED	True

3. Semantic projection and coherent governance loading

This section explains why the governed state remained ESCALATE while preserving the Decision API as authoritative.

Projection status	COMPLETE
Selected projection	ESCALATE
Decision API action	ESCALATE
Projection disagreement	True
Disagreement detail	{"decision_api_action_preserved": "ESCALATE", "highest_scoring_projection": "ALLOW", "override_performed": false}
State scores	{"ALLOW": 0.69237932, "BLOCK": 0.25912270000000004, "ESCALATE": 0.5498677000000001}

Why this matters

Projection is explanatory only. It can surface tension between governance signals, but it does not override release control. The authoritative Decision API preserved ESCALATE and withheld autonomous release.

Coherent Governance Loading

Loading score	0.574282
Loading tier	PARTIAL
Component scores	{"benchmark_alignment": 0.0, "evidence_coverage": 0.0, "policy_alignment": 1.0, "risk_compatibility": 1.0, "trace_density": 0.9, "trust_support": 0.8149, "uncertainty_compatibility": 0.32999999999999996}
Missing components	evidence_pack

4. Evidence geodesic, observer views, and packet projection budget

Evidence geodesic status	COMPLETE
Geodesic path count	1
Distance formula	$\text{distance} = -\log(\max(\text{semantic_entanglement}, 1e-6))$
Governance path ensemble status	COMPLETE
Path count	5

Observer views

Role	Question	Answer
buyer_executive	Can this AI output move autonomously?	Governed by Decision API action: ESCALATE
human_reviewer	What must be reviewed next?	Review Decision API reasons, evidence, pending evidence, and projection disagreement if present.
auditor	What proves the governance path?	Trace, evidence pack, route proof, hashes, and semantic projection provenance.

Packet projection budget

Buyer explanation	The PDF renders a decision-relevant subset; full_audit.governance.json preserves the complete canonical record.
Max evidence items rendered	12
Max trace nodes rendered	20
Max trace links rendered	30
Canonical JSON preserves full record	True

5. Human review and improvement loop

Human review status	REQUIRED
Review owner role	None
Review queue	compliance_risk_review
Review options	APPROVE, REVISE, BLOCK, REPLACE_WITH_SAFE_TEMPLATE, REQUEST_MORE_EVIDENCE
Outcome closure	PENDING
L3 signal export	PENDING_REVIEW
Improvement signal status	REVIEW_GATED
Training signal export	PENDING_REVIEW

Public interpretation

- The current packet routes the output to human review and preserves the outcome as pending.
- Improvement signals are review-gated; the original packet remains immutable.
- Historical/trace run features are safely marked insufficient history until comparable accepted packets exist.

Final limitation statement

This public kit demonstrates controlled-harness runtime-seam proof and semantic explanation. Production non-bypassability, customer-specific baselines, and live customer route enforcement require attached production integration evidence and are not asserted here.